	Course Name: Advanced Web Technology	EXPERIMENT NO. 4
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Building models with Mongoose

Create a database schema and model using mongoose and perform MongoDB queries for operations such as insert, update, and delete while also establishing a connection to the MongoDB database.

Mongoose Connection

```
const mongoose = require("mongoose");

mongoose
  .connect("mongodb://localhost/mydb")
  .then(() => console.log("Connected to MongoDB successfully!"))
  .catch((error) => console.error("Error connecting to MongoDB", error));
```

```
(base) PS D:\DEV\Web Development\PDEU-AWT\Practical_4> npm start

> practical_4@1.0.0 start
> nodemon task1.js

[nodemon] 3.1.0
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node task1.js`
Connected to MongoDB successfully!
```

Database Schema

```
const schema = new mongoose.Schema({

  name: {
    type: String,
    required: true,
  },
  email: {
    type: String,
```

```
    required: true,
    unique: true,
  },
  age: {
    type: Number,
    default: 0,
  },
  deregistered: {
    type: Date,
    default: Date.now,
  },
});
const obj = mongoose.model("subsription", schema);
```

Insert Operation

```
const newobj = new obj({
  name: "Harsh Shah",
  email: "harsh.sce21@sot.pdpu.ac.in",
  age: 20,
});
newobj
  .save()
  .then((kri) => console.log("User created:", kri))
  .catch((err) => console.error(err));
```

```
Connected to MongoDB successfully!
User created: {
  name: 'Harsh Shah',
  email: 'harsh.sce21@sot.pdpu.ac.in',
  age: 20,
  _id: new ObjectId('660fd63ed60d5b00ccc41371'),
  deregistered: 2024-04-05T10:45:18.554Z,
  __v: 0
}
```

```
_id: ObjectId('660fd63ed60d5b00ccc41371')
name : "Harsh Shah"
email : "harsh.sce21@sot.pdpu.ac.in"
age : 20
deregistered : 2024-04-05T10:45:18.554+00:00
__v : 0
```

a. Insert One

```
const Newuser = new obj({ name: "krish", email: "ky@gmail.com", age: 20 });
```

```
obj.collection
```

```
.insertOne(Newuser)
```

```
.then((data) => {
```

```
  console.log(data);
```

```
})
```

```
.catch((err) => {
```

```
  console.log(err);
```

```
});
```

```
Connected to MongoDB successfully!
{
  acknowledged: true,
  insertedId: new ObjectId('660fd845ddcedbff28571b2e')
}
```

b. Insert Many

```
const Newuser = new obj({ name: "krishna", email: "ku@gmail.com", age: 20 });
```

```
const Newuser1 = new obj({ name: "kriti", email: "kr@gmail.com", age: 21 });
```

```
obj.collection
```

```
.insertMany([Newuser, Newuser1])
```

```
.then((data) => {
```

```
  console.log(data);
```

```
})
```

```
.catch((err) => {
```

```
  console.log(err);
```

```
});
```

```
Connected to MongoDB successfully!
{
  acknowledged: true,
  insertedCount: 2,
  insertedIds: {
    '0': new ObjectId('660fd8829d209dcbb3368d5b'),
    '1': new ObjectId('660fd8829d209dcbb3368d5c')
  }
}
```

Update Operation

a. Update One

```
obj.updateOne({ name: "krishna" }, { $set: { name: "madhav" } })  
  .then((docs) => {  
    if (docs) {  
      console.log(docs);  
    } else {  
      console.log("no such user exist");  
    }  
  })  
  .catch((err) => {  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!  
{  
  acknowledged: true,  
  modifiedCount: 1,  
  upsertedId: null,  
  upsertedCount: 0,  
  matchedCount: 1  
}
```

b. Update Many

```
obj.updateMany({age: 20},{ $set:{ age: 15}})  
  .then((docs)=>{  
    if(docs) {  
      console.log(docs);  
    } else {  
      console.log("no such user exist");  
    }  
  })  
  .catch((err)=>{  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!  
{  
  acknowledged: true,  
  modifiedCount: 4,  
  upsertedId: null,  
  upsertedCount: 0,  
  matchedCount: 4  
}
```

Find/Search Operation

a. Find without query

```
obj.find({})  
  
  .then((doc) => {  
    console.log(doc);  
  })  
  
  .catch((err) => {  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!  
[  
  {  
    _id: new ObjectId('660fd63ed60d5b00ccc41371'),  
    name: 'Harsh Shah',  
    email: 'harsh.sce21@sot.pdpu.ac.in',  
    age: 15,  
    deregistered: 2024-04-05T10:45:18.554Z,  
    __v: 0  
  },  
  {  
    _id: new ObjectId('660fd845ddcedbff28571b2e'),  
    name: 'krish',  
    email: 'ky@gmail.com',  
    age: 15,  
    deregistered: 2024-04-05T10:53:57.063Z  
  },  
  {  
    _id: new ObjectId('660fd8829d209dcb3368d5b'),  
    name: 'madhav',  
    email: 'ku@gmail.com',  
    age: 15,  
    deregistered: 2024-04-05T10:54:58.784Z  
  },  
  {  
    _id: new ObjectId('660fd8829d209dcb3368d5c'),  
    name: 'kriti',  
    email: 'kr@gmail.com',  
    age: 21,  
    deregistered: 2024-04-05T10:54:58.784Z  
  },  
  {  
    _id: new ObjectId('660fd92a27c646b3dd442ec0'),  
    name: 'dev',  
    email: 'dev@gmail.com',  
    age: 15,  
    deregistered: 2024-04-05T10:57:46.699Z  
  }  
]
```

b. Find with query

```
obj.find({ name: "dev" })  
  
  .then((doc) => {  
    console.log(doc);  
  })  
  
  .catch((err) => {  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!
[
  {
    _id: new ObjectId('660fd92a27c646b3dd442ec0'),
    name: 'dev',
    email: 'dev@gmail.com',
    age: 15,
    deregistered: 2024-04-05T10:57:46.699Z
  }
]
```

Delete Operation

a. Delete One

```
obj.deleteOne({name: "kriti"})
.then((docs)=>{
  if(docs) {
    console.log(docs);
  } else {
    console.log("no such user exist");
  }
}).catch((err)=>{
  console.log(err);
});
```

```
Connected to MongoDB successfully!
{ acknowledged: true, deletedCount: 1 }
```

b. Delete Many

```
obj.deleteMany({age: 21})
.then((docs)=>{
  if(docs) {
    console.log(docs);
  } else {
    console.log("no such user exist");
  }
}).catch((err)=>{
  console.log(err);
});
```

```
Connected to MongoDB successfully!
{ acknowledged: true, deletedCount: 4 }
```

MongoDB Compass

mydb.subscriptions

4DOCUMENTS2INDEXES

Documents

Aggregations

Schema

Indexes

Validation

Filter

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

</>

Options

ADD DATA

EXPORT DATA

UPDATE

DELETE

1 - 4 of 4

```
{
  "_id": ObjectId('660fd63ed60d5b00ccc41371'),
  "name": "Harsh Shah",
  "email": "harsh.sce21@sot.pdpu.ac.in",
  "age": 15,
  "deregistered": 2024-04-05T10:45:18.554+00:00,
  "__v": 0
}
```

```
{
  "_id": ObjectId('660fd845ddcedbff28571b2e'),
  "name": "krish",
  "email": "ky@gmail.com",
  "age": 15,
  "deregistered": 2024-04-05T10:53:57.063+00:00
}
```

```
{
  "_id": ObjectId('660fd8829d209dcbb3368d5b'),
  "name": "madhav",
  "email": "ku@gmail.com",
  "age": 15,
  "deregistered": 2024-04-05T10:54:58.784+00:00
}
```

My Queries

mydb

mydb

subscriptions

mydb.subscriptions

0DOCUMENTS2INDEXES

Documents

Aggregations

Schema

Indexes

Validation

Filter

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

</>

Options

ADD DATA

EXPORT DATA

UPDATE

DELETE

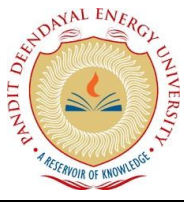
0 - 0 of 0



This collection has no data

It only takes a few seconds to import data from a JSON or CSV file.

Import Data

	Course Name: Advanced Web Technology	EXPERIMENT NO. 5
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Create your first React code

1. Create a “hello world” react app.

index.js

```
import React from "react";
import ReactDOM from "react-dom/client";
import "./index.css";
import App from "./App";
import reportWebVitals from "./reportWebVitals";

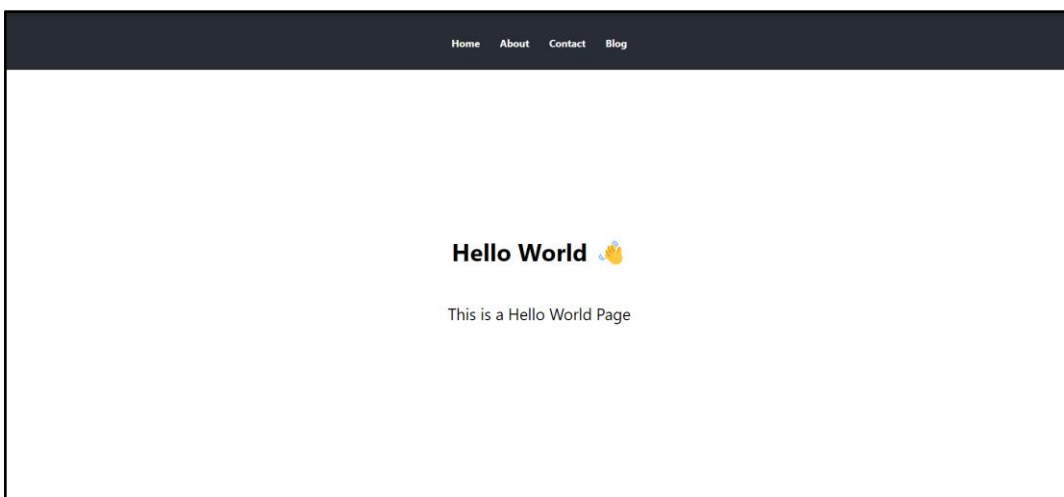
const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
reportWebVitals();
```

App.js

```
import "./App.css";
import Home from "./components/Home";
import { BrowserRouter, Routes, Route } from "react-router-dom";
function App() {
  return (
    <div>
      <BrowserRouter>
        <Routes>
          <Route path="/" element={<Home />}>
            {" "}
          </Route>
        </Routes>
      </BrowserRouter>
    </div>
  );
}
export default App;
```

Home.jsx

```
import React from "react";
const Home = () => {
  return
    <div>
      <header className="App-header">
        <h2>Hello World 🙋‍♂️ </h2>
        <p>This is a Hello World Page</p>
      </header>
    </div>
  );
};
export default Home;
```

Output**2. Create a multiple web page commercial furniture application having a navigation bar using react components having pages such as home, products and customer review.****index.js**

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import './index.css';
import './App.css';
import App from './App';
import reportWebVitals from './reportWebVitals';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <App/>
  </React.StrictMode>
);
reportWebVitals();
```

App.js

```
import './App.css';
import NavBar from './components/NavBar';
import Home from './components/Home';
import Products from './components/Products';
import Review from './components/Review';
import { BrowserRouter, Routes, Route } from 'react-router-dom';

function App() {
  return (
    <div>
      <BrowserRouter>
        <NavBar />
        <Routes>
          <Route path="/" element={<Home />}>
            {" "}
          </Route>
          <Route path="/reviews" element={<Review />}>
            {" "}
          </Route>
          <Route path="/products" element={<Products />}>
            {" "}
          </Route>
        </Routes>
      </BrowserRouter>
    </div>
  );
}
export default App;
```

Home.jsx

```
import React from 'react';
import './Home.module.css';

const Home = () => {
  return (
    <div>
      <header className="App-header">
        <div className="home">
          <div className="hero-section">
            <h1>Welcome to Our Commercial Furniture Store</h1>
            <p>
              We provide high-quality furniture solutions for
              commercial spaces.
            </p>
          </div>
          <div className="about-section">
            <h2>About Us</h2>
            <p>
              We are dedicated to providing durable and stylish
              furniture options for businesses, offices,
              restaurants, hotels, and more.
            </p>
            <p>
              Our team is committed to delivering exceptional
```

```

        customer service and ensuring satisfaction with
        every purchase.
    </p>
</div>
<div className="contact-section">
    <h2>Contact Us</h2>
    <ul>
        <li>Email: info@furniturestore.com</li>
        <li>Phone: 123-456-7890</li>
    </ul>
</div>
</div>
</header>
</div>
);
};
export default Home;

```

Home.module.css

```

.home {
    padding: 20px;
}

.hero-section {
    text-align: center;
    margin-bottom: 50px;
}

.hero-section h1 {
    font-size: 36px;
    margin-bottom: 20px;
}

.hero-section p {
    font-size: 18px;
    margin-bottom: 30px;
}

.btn {
    display: inline-block;
    padding: 10px 20px;
    background-color: #007bff;
    color: #fff;
    text-decoration: none;
    border-radius: 5px;
    transition: background-color 0.3s ease;
}

.btn:hover {
    background-color: #0056b3;
}

.about-section,
.contact-section {
    margin-bottom: 50px;
}

.about-section h2,
.contact-section h2 {
    font-size: 24px;
    margin-bottom: 20px;
}

.about-section p,

```

```
.contact-section p {
  font-size: 16px;
  margin-bottom: 15px;
}
.contact-section ul {
  list-style: none;
  padding-left: 0;
}
.contact-section li {
  font-size: 16px;
  margin-bottom: 10px;
}
```

NavBar.jsx

```
import React from "react";
import { Link } from "react-router-dom";
import styles from "./NavBar.module.css";
const NavBar = () => {
  return (
    <nav className={styles.navBar}>
      <ul className={styles.navList}>
        <li>
          <Link to="/">Home</Link>
        </li>
        <li>
          <Link to="/products">Products</Link>
        </li>
        <li>
          <Link to="/reviews">Reviews</Link>
        </li>
      </ul>
    </nav>
  );
};
export default NavBar;
```

NavBar.module.css

```
.navBar {
  background-color: #282c34;
  padding: 1em;
}
.navList {
  list-style-type: none;
  padding: 0;
  display: flex;
  justify-content: center;
}
.navList li {
  margin: 0 1em;
}
.navList li a {
  text-decoration: none;
  color: white;
  font-weight: bold;
}
```

Furniture.js

```
import "../App.css";
function Furniture(props) {
  return (
    <div className="grid-item">
      
      <h3 className="itemName">Trotten</h3>
      <p className="itemPrice">16690</p>
    </div>
  );
}
export default Furniture;
```

Products.jsx

```
import "../App.css";
import Furniture from "./Furniture";

function Products() {
  return (
    <div className="grid-container App-header App">
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
    </div>
  );
}

export default Products;
```

Review.jsx

```
import React from "react";
import "../App.css";
const reviews = [
  {
    id: 1,
    name: "John Doe",
    rating: 5,
    review: "Excellent product! Very satisfied with the quality and performance.",
  },
  {
    id: 2,
    name: "Jane Smith",
    rating: 4,
```

```

    review: "Great service! Fast delivery and good customer support.",
  },
  {
    id: 3,
    name: "David Brown",
    rating: 3,
    review: "Product was okay, but could be improved. Delivery was delayed.",
  },
];

```

```

const Reviews = ({ name, rating, review }) => {
  return (
    <div className="review">
      <h3>{name}</h3>
      <div className="rating">Rating: {rating}/5</div>
      <p>{review}</p>
    </div>
  );
};

```

```

const CustomerReviews = ({ reviews }) => {
  return (
    <div className="customer-reviews">
      <h2>Customer Reviews</h2>
      {reviews.map((review) => (
        <Reviews key={review.id} {...review} />
      ))}
    </div>
  );
};

```

```

const Review = () => {
  return (
    <div>
      <header className="App-header">
        <CustomerReviews reviews={reviews} />
      </header>
    </div>
  );
};

```

```
export default Review;
```

App.css

```

.App {
  text-align: center;
  width: 97.5vw;
}
.App-logo {
  height: 40vmin;
  pointer-events: none;
}
@media (prefers-reduced-motion: no-preference) {
  .App-logo {
    animation: App-logo-spin infinite 20s linear;
  }
}

```

```
.App-header {
  background-color: #282c34;
  min-height: 100vh;
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  font-size: calc(10px + 2vmin);
  color: white;
}
.App-link {
  color: #61dafb;
}
@keyframes App-logo-spin {
  from {
    transform: rotate(0deg);
  }
  to {
    transform: rotate(360deg);
  }
}
.link {
  color: white;
}
.grid-container {
  display: inline-grid;
  grid-template-columns: auto auto auto;
  grid-gap: 20px;
  text-align: center;
  padding: 20px;
}
.grid-item {
  background-color: #f2f2f2;
  padding: 20px;
  text-align: center;
  color: black;
  border-radius: 25px;
}
.customer-reviews {
  margin: 20px auto;
  width: 70%;
}
.review {
  border: 1px solid #ccc;
  border-radius: 8px;
  padding: 20px;
  margin-bottom: 20px;
  box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
}
.rating {
  font-weight: bold;
  margin-bottom: 10px;
}
```

Output:

Home Page

HomeProductsReviews

Welcome to Our Commercial Furniture Store

We provide high-quality furniture solutions for commercial spaces.

About Us

We are dedicated to providing durable and stylish furniture options for businesses, offices, restaurants, hotels, and more.

Our team is committed to delivering exceptional customer service and ensuring satisfaction with every purchase.

Contact Us

Email: info@furniturestore.com

Phone: 123-456-7890

Products Page

HomeProductsReviews



Trotten

16690



Trotten

16690



Trotten

16690



Trotten

16690



Trotten

16690



Trotten

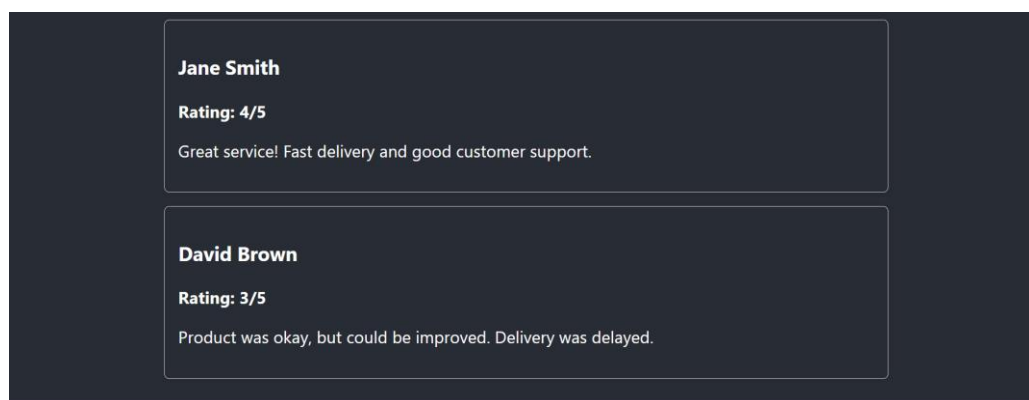
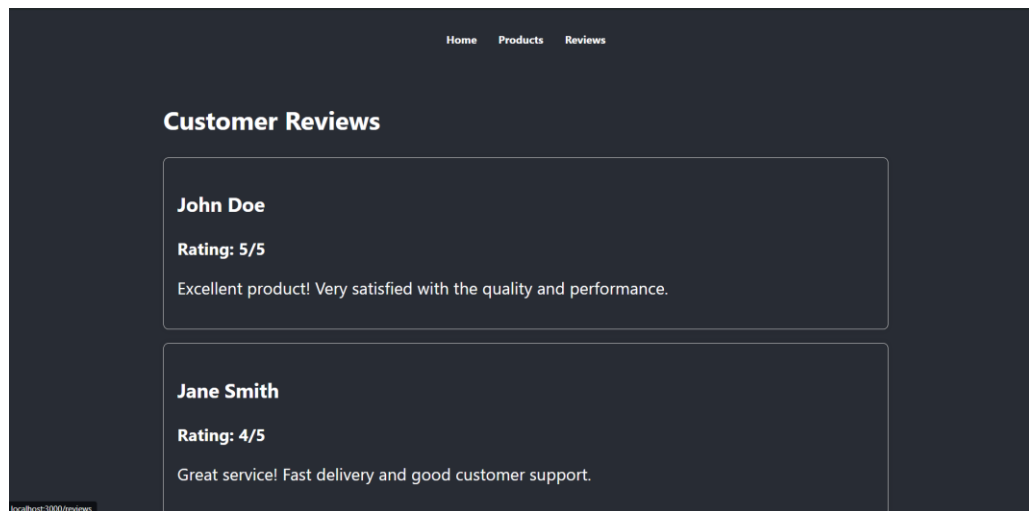
16690



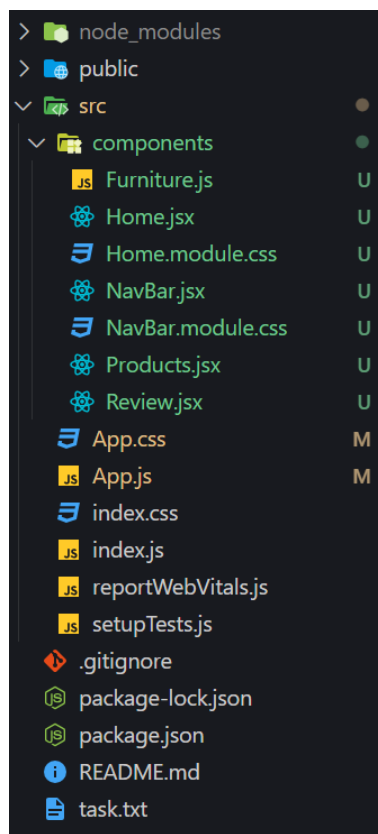


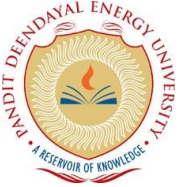


Customer Review page



Folder Structure



	Course Name: Advanced Web Technology	EXPERIMENT NO. 6
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Working with properties in React

1. In the commercial furniture web-application add products with information like image, price, color, manufacturer using react props.

App.js

```
import './App.css';
import Home from './Home';
import { BrowserRouter, Routes, Route } from 'react-router-dom';
import Task1and2 from './Task1and2/Task1and2';
import Task3and4 from './Task3and4/Task3and4';
function App() {
  return (
    <div>
      <BrowserRouter>
        <Routes>
          <Route path="/" element={ <Home /> } > </Route>
          <Route path="/student" element={ <Task1and2 /> } > </Route>
          <Route path="/furniture" element={ <Task3and4 /> } > </Route>
        </Routes>
      </BrowserRouter>
    </div>
  );
}
export default App;
```

Home.jsx

```
import './App.css';
import React from 'react'
import { Link } from 'react-router-dom';
const Home = () => {
  return (
    <div>
      <header className="App-header" style={{ width: '100vw' }}>
        <h2>Practical 6</h2>
        <Link to="/student" className="link">Go to Task 1 and 2</Link>
        <br />
        <Link to="/furniture" className="link">Go to Task 3 and 4</Link>
      </header>
    </div>
  )
}
export default Home;
```

Task3and4.js

```
import "../App.css";
import Furniture from "../Furniture";

function Task3and4() {
  return (
    <div className="grid-container App-header App">
      <Furniture
        itemName="Linanas"
        itemPrice="29990"
        itemImg="https://www.ikea.com/in/en/images/products/linanaes-3-seat-sofa-with-chaise-longue-vissle-
dark-grey__1013909_pe829463_s5.jpg?f=xxs"
        color="Gray"
        manufacturer="Ravi Furnitures"
      />

      <Furniture
        itemName="Nodeland"
        itemPrice="15990"
        itemImg="https://www.ikea.com/in/en/images/products/nodeland-wardrobe-with-3-doors-medium-
brown__0836934_pe778612_s5.jpg?f=xxs"
        color="Brown"
        manufacturer="New india Furnitures"
      />

      <Furniture
        itemName="Nolmyra"
        itemPrice="3990"
        itemImg="https://www.ikea.com/in/en/images/products/renberget-swivel-chair-bomstad-
black__1025981_pe834276_s5.jpg?f=xxs"
        color="Black"
        manufacturer="IKEA"
      />

      <Furniture
        itemName="Oxberg"
        itemPrice="3690"
        itemImg="https://www.ikea.com/in/en/images/products/billy-oxberg-bookcase-with-door-
white__1094849_pe863651_s5.jpg?f=xxs"
        color="White"
        manufacturer="he Interior Hub"
      />

      <Furniture
        itemName="Knarrevik"
        itemPrice="899"
        itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs"
        color="Black"
        manufacturer="Furniture Nest"
      />

      <Furniture
        itemName="Tufjord"
        itemPrice="42990"
        itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs"
```

```
        color="Black"
        manufacturer="IKEA"
    />
</div>
);
}
export default Task3and4;
```

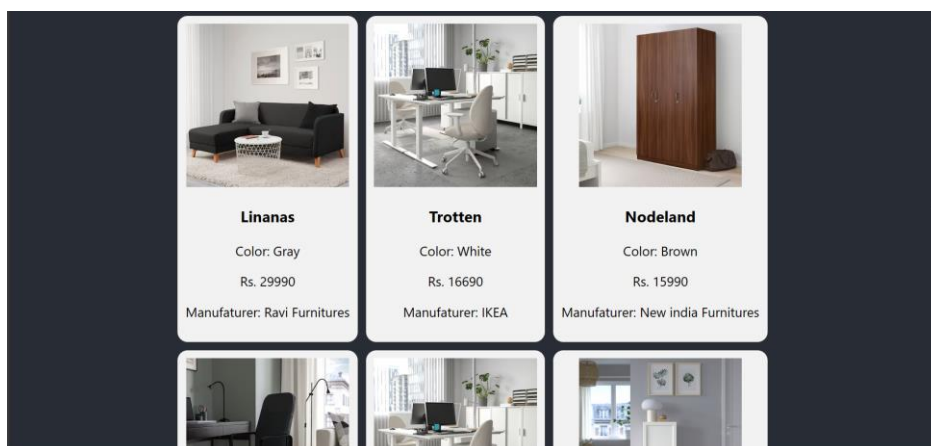
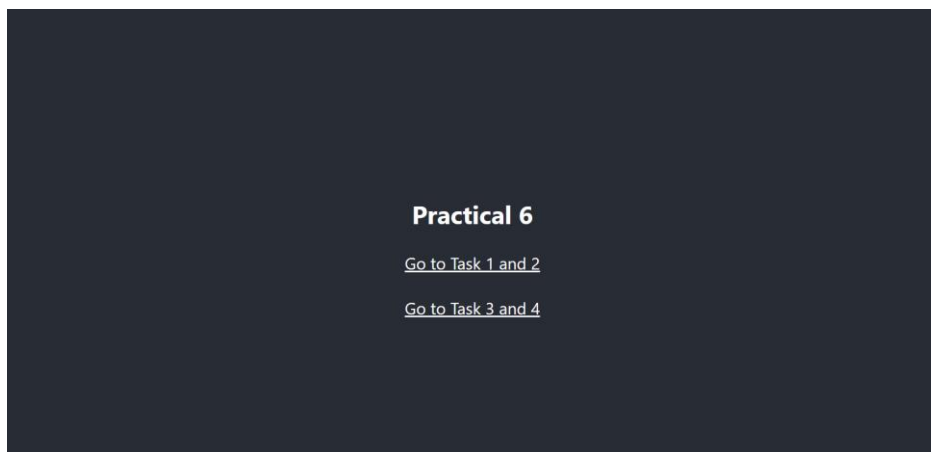
Furniture.js

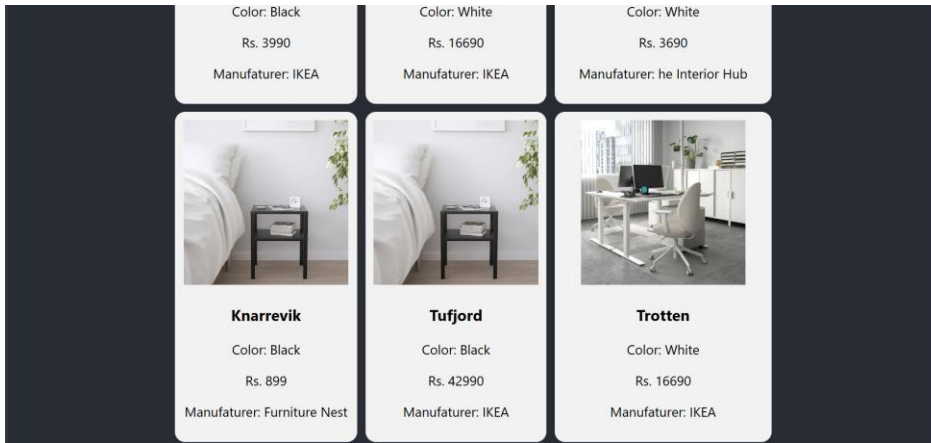
```
import "../App.css";

function Furniture(props) {
    return (
        <div className="grid-item">
            <img src={props.itemImg} alt={props.itemName} className="itemImg" />
            <h3 className="itemName">{props.itemName}</h3>
            <p className="itemPrice">Color: {props.color}</p>
            <p className="itemPrice">Rs. {props.itemPrice}</p>
            <p className="itemPrice">Manufaturer: {props.manufacturer}</p>
        </div>
    );
}
"IKEA",
};

export default Furniture;
```

Output





2. Add default values for all the furniture properties which are displayed in absence of corresponding information.

Modified Task3and4.js

```
import "../App.css";
import Furniture from "../Furniture";

function Task3and4() {
  return (
    <div className="grid-container App-header App">
      <Furniture
        itemName="Linanas"
        itemPrice="29990"
        itemImg="https://www.ikea.com/in/en/images/products/linanaes-3-seat-sofa-with-chaise-longue-vissle-dark-grey__1013909_pe829463_s5.jpg?f=xxs"
        color="Gray"
        manufacturer="Ravi Furnitures"
      />

      <Furniture />

      <Furniture
        itemName="Nodeland"
        itemPrice="15990"
        itemImg="https://www.ikea.com/in/en/images/products/nodeland-wardrobe-with-3-doors-medium-brown__0836934_pe778612_s5.jpg?f=xxs"
        color="Brown"
        manufacturer="New india Furnitures"
      />

      <Furniture
        itemName="Nolmyra"
        itemPrice="3990"
        itemImg="https://www.ikea.com/in/en/images/products/renberget-swivel-chair-bomstad-black__1025981_pe834276_s5.jpg?f=xxs"
        color="Black"
        manufacturer="IKEA"
      />

      <Furniture />
    </div>
  );
}
```

```

    <Furniture
      itemName="Oxberg"
      itemPrice="3690"
      itemImg="https://www.ikea.com/in/en/images/products/billy-oxberg-bookcase-with-door-
white__1094849_pe863651_s5.jpg?f=xxs"
      color="White"
      manufacturer="The Interior Hub"
    />

    <Furniture
      itemName="Knarrevik"
      itemPrice="899"
      itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs"
      color="Black"
      manufacturer="Furniture Nest"
    />

    <Furniture
      itemName="Tufjord"
      itemPrice="42990"
      itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs"
      color="Black"
      manufacturer="IKEA"
    />

    <Furniture />
  </div>
);
}
export default Task3and4;

```

Modified Furniture.js

```

import "../App.css";

function Furniture(props) {
  return (
    <div className="grid-item">
      <img src={props.itemImg} alt={props.itemName} className="itemImg" />
      <h3 className="itemName">{props.itemName}</h3>
      <p className="itemPrice">Color: {props.color}</p>
      <p className="itemPrice">Rs. {props.itemPrice}</p>
      <p className="itemPrice">Manufacturer: {props.manufacturer}</p>
    </div>
  );
}

Furniture.defaultProps = {
  itemName: "Trotten",
  itemPrice: "16690",
  itemImg:
    "https://www.ikea.com/in/en/images/products/trotten-desk-sit-stand-white__1040109_pe840587_s5.jpg?f=xxs",
  color: "White",
  manufacturer: "IKEA",
};
export default Furniture;

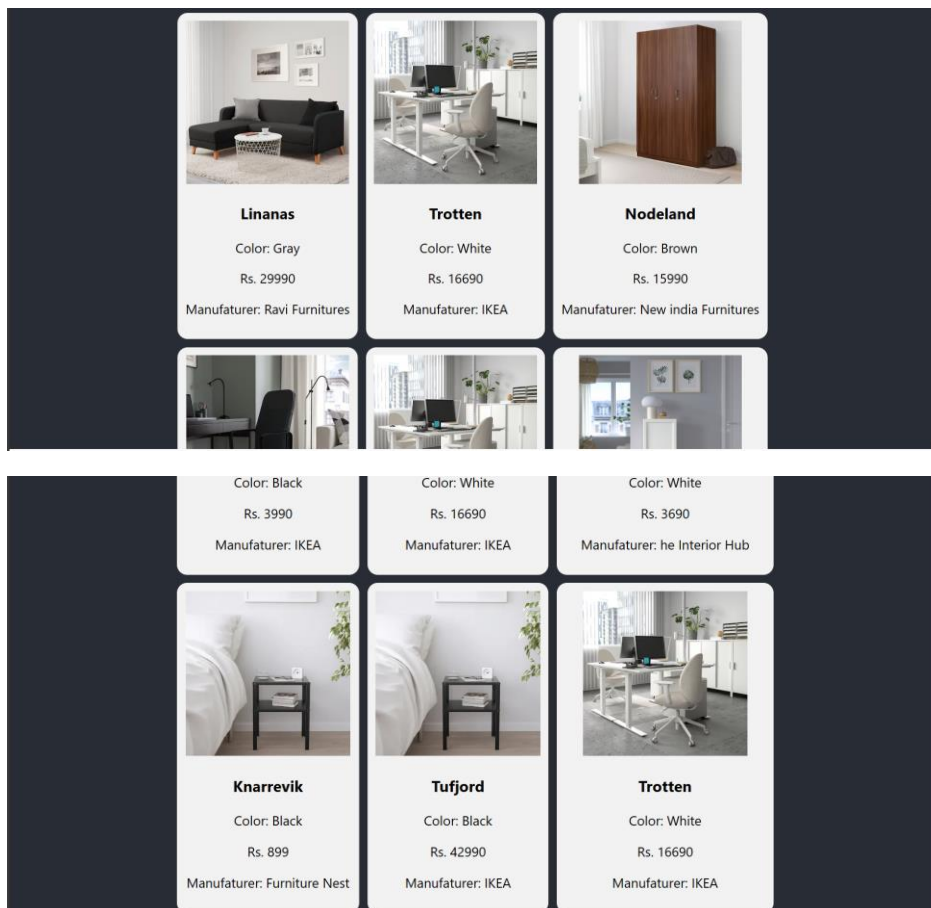
```

Output

Practical 6

[Go to Task 1 and 2](#)

[Go to Task 3 and 4](#)

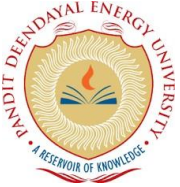


You can now view practical-6-1 in the browser.

Local: `http://localhost:3000`
On Your Network: `http://172.29.32.1:3000`

Note that the development build is not optimized.
To create a production build, use `npm run build`.

webpack compiled **successfully**

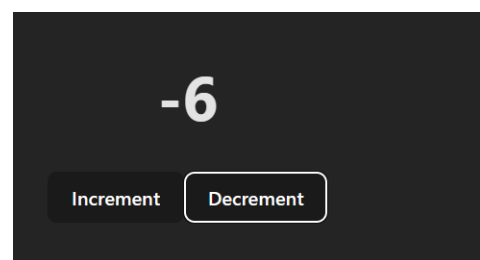
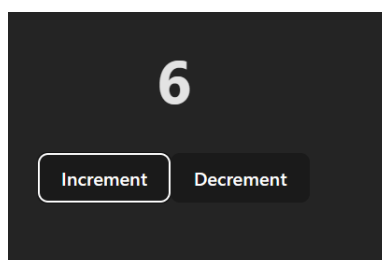
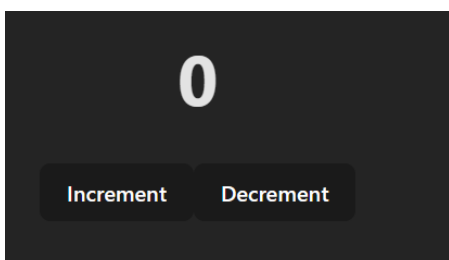
	Course Name: Advanced Web Technology	EXPERIMENT NO. 7
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Working with states in React

1. Create a counter based on react states using both the approaches class component as well as functional component.

Approach 1: Class Component

```
import React from "react";
class App extends React.Component {
  constructor(counter) {
    super(counter);
    this.state = {
      counter: 0,
    };
  }
  increment = () => {
    this.setState((prevState) => ({
      counter: prevState.counter + 1
    }));
  }
  decrement = () => {
    this.setState((prevState) => ({
      counter: prevState.counter - 1
    }));
  }
  render() {
    return (
      <div>
        <h1>{this.state.counter}</h1>
        <button onClick={this.increment}>Increment</button>
        <button onClick={this.decrement}>Decrement</button>
      </div>
    );
  }
}
export default App;
```

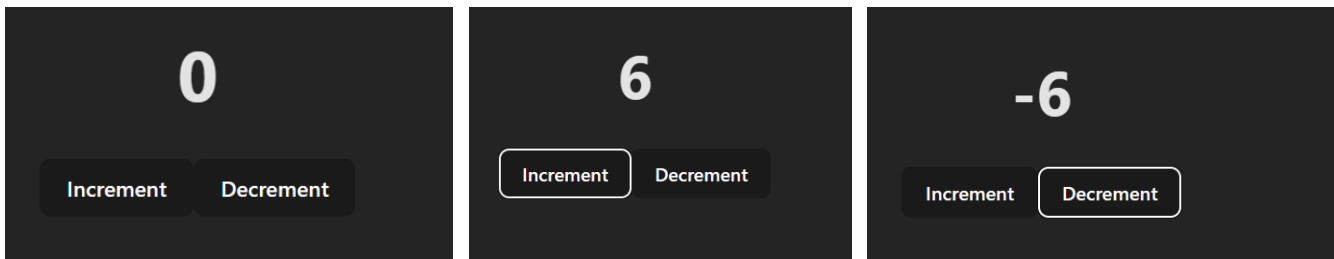


Approach 2: Functional Component

```
import { useState } from "react";
function App2() {
  const [count, setCount] = useState(0);

  const increment = () => {
    setCount(count + 1);
  };

  const decrement = () => {
    setCount(count - 1);
  };
  return (
    <div>
      <h1>{count}</h1>
      <button onClick={increment}>Increment</button>
      <button onClick={decrement}>Decrement</button>
    </div>
  );
}
export default App2;
```

**2. In the commercial furniture web-application add a like button under each product in the products page using react states.**

```
import Furniture from './Furniture';

function App3() {
  return (
    <div className="grid-container App-header App">
      <Furniture itemName="Linanas" itemPrice="29990"
itemImg="https://www.ikea.com/in/en/images/products/linanaes-3-seat-sofa-with-chaise-longue-vissle-dark-
grey__1013909_pe829463_s5.jpg?f=xxs" />

      <Furniture />

      <Furniture itemName="Nodeland" itemPrice="15990"
itemImg="https://www.ikea.com/in/en/images/products/nodeland-wardrobe-with-3-doors-medium-
brown__0836934_pe778612_s5.jpg?f=xxs" />

      <Furniture itemName="Nolmyra" itemPrice="3990"
itemImg="https://www.ikea.com/in/en/images/products/renberget-swivel-chair-bomstad-
black__1025981_pe834276_s5.jpg?f=xxs" />

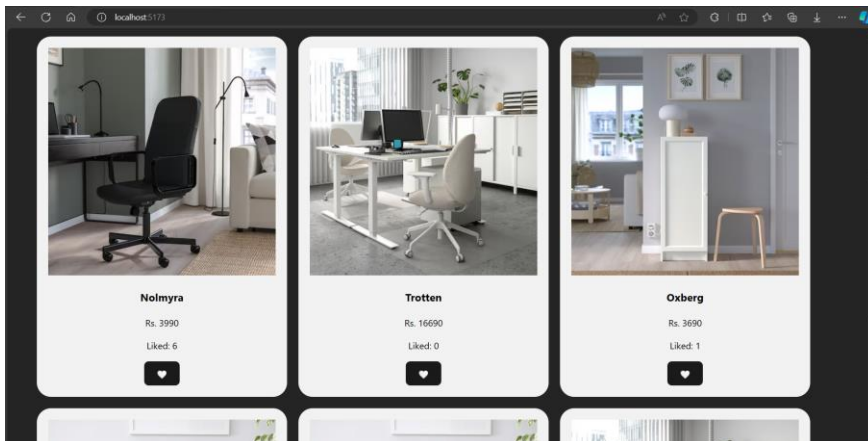
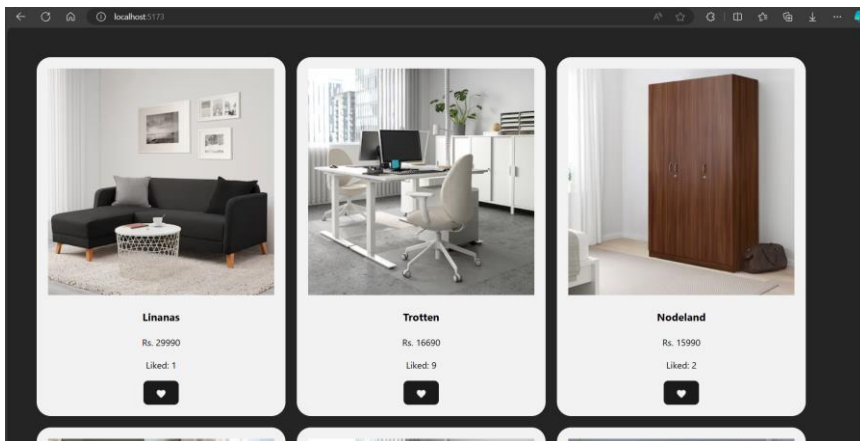
      <Furniture />
    </div>
  );
}
```

```
<Furniture itemName="Knarrevik" itemPrice="899"
itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs" />
```

```
<Furniture itemName="Tufjord" itemPrice="42990"
itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs" />
```

```
</div>
);
}
export default App3;
```

Output



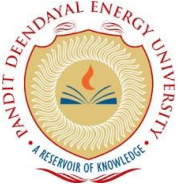
```
PowerShell 7.4.1
PS D:\DEV\Web Development\PDEU-AWT\Practical_7\practical-7-1> npm run dev

> practical-7-1@0.0.0 dev
> vite

Re-optimizing dependencies because vite config has changed

VITE v5.1.5 ready in 485 ms

➔ Local:   http://localhost:5173/
➔ Network: use --host to expose
➔ press h + enter to show help
11:44:41 am [vite] page reload src/main.jsx
11:44:51 am [vite] page reload src/main.jsx (x2)
11:50:59 am [vite] page reload src/main.jsx (x3)
11:51:37 am [vite] hmr update /src/Furniture.jsx
11:52:24 am [vite] hmr update /src/Furniture.jsx (x2)
11:52:27 am [vite] hmr update /src/Furniture.jsx (x3)
11:53:48 am [vite] hmr update /src/Furniture.jsx (x4)
11:54:00 am [vite] hmr update /src/Furniture.jsx (x5)
```

	Course Name: Advanced Web Technology	EXPERIMENT NO. 8
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Working with forms in React

1. Create a customer review form using react forms for the commercial furniture web-application with multiple fields such as name of the customer(text), the ratings for the customer experience out of 5(a drop-down menu), whether interested in buying again in future (check-box yes or no) and message from the customer(text-area) also ensure that the message should be less than 25 characters.

index.js

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App.jsx";
import "./index.css";
import { BrowserRouter } from "react-router-dom";

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </React.StrictMode>
);
```

App.js

```
import { Route, Navigate, Routes } from "react-router-dom";
import CustomerReviewForm from "./CustomerReviewForm";
import ThankYouPage from "./ThankyouPage";

function App() {
  return (
    <div>
      <Routes>
        <Route path="/" element={<CustomerReviewForm />} />
        <Route path="/thankyou" element={<ThankYouPage />} />
        <Route path="*" element={<Navigate to="/" />}></Route>
      </Routes>
    </div>
  );
}
export default App;
```

CustomerReviewForm.jsx

```

import { useEffect, useState } from "react";
import { useNavigate } from "react-router-dom";

const CustomerReviewForm = () => {
  const navigate = useNavigate();
  const [submitted, setSubmitted] = useState(false);
  const [name, setName] = useState("");
  const [rating, setRating] = useState("");
  const [interested, setInterested] = useState(false);
  const [message, setMessage] = useState("");

  const submitForm = (e) => {
    e.preventDefault();
    setSubmitted(true);
  };

  useEffect(() => {
    if (submitted) {
      console.log("Form submitted: ", name, rating, interested, message);
      navigate("/thankyou", {
        replace: true,
        state: { name, rating, interested, message },
      });
    }
  }, [submitted, name, rating, interested, message, navigate]);

  return (
    <div>
      <h1>Commercial Furniture Web Application</h1>
      <form onSubmit={(e) => submitForm(e)}>
        <table>
          <tbody>
            <tr>
              <td>
                <label>Name:</label>
              </td>
              <td>
                <input
                  type="text"
                  value={name}
                  onChange={(e) => setName(e.target.value)}
                />
              </td>
            </tr>
            <tr>
              <td>
                <label>Rating:</label>
              </td>
              <td>
                <select
                  value={rating}
                  onChange={(e) => setRating(e.target.value)}
                >
                  <option value="">Select</option>
                  <option value="1">1</option>
                  <option value="2">2</option>
                </select>
              </td>
            </tr>
          </tbody>
        </table>
      </form>
    </div>
  );
};

```

```

        <option value="3">3</option>
        <option value="4">4</option>
        <option value="5">5</option>
    </select>
</td>
</tr>
<tr>
<td>
    <label>
        Interested in buying again in future:
    </label>
</td>
<td>
    <label className="checkbox-label">
        <input
            type="checkbox"
            checked={interested}
            onChange={(e) =>
                setInterested(e.target.checked)
            }
        />
        Yes
    </label>
</td>
</tr>
<tr>
<td>
    <label>Message (Max 25 characters):</label>
</td>
<td>
    <textarea
        value={message}
        onChange={(e) => setMessage(e.target.value)}
        maxLength={25}
    />
</td>
</tr>
<tr>
<td colspan="2">
    <button type="submit">Submit</button>
</td>
</tr>
</tbody>
</table>
</form>
</div>
);
};
export default CustomerReviewForm;

```

ThankyouPage.jsx

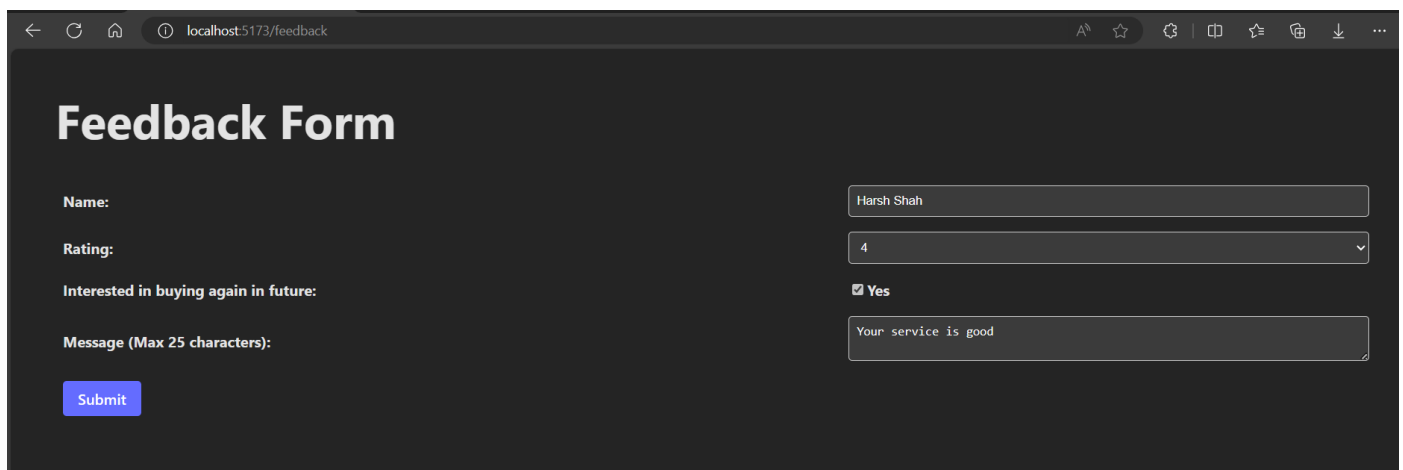
```
import { useLocation } from "react-router-dom";

const ThankYouPage = () => {
  const location = useLocation();
  const formData = location.state;

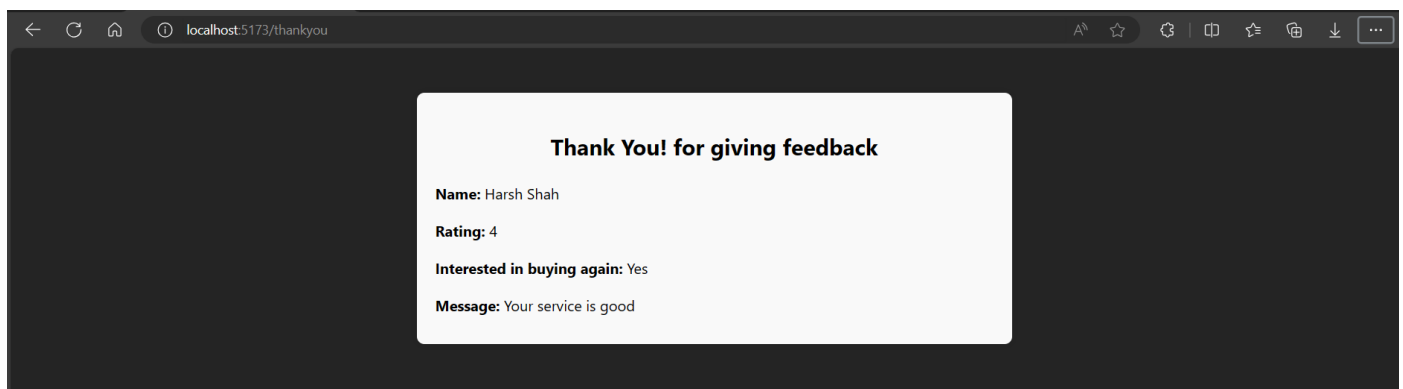
  return (
    <div>
      <h2>Thank You! for giving feedback</h2>
      <p>Name: {formData.name}</p>
      <p>Rating: {formData.rating}</p>
      <p>
        Interested in buying again: {formData.interested ? "Yes" : "No"}
      </p>
      <p>Message: {formData.message}</p>
    </div>
  );
};

export default ThankYouPage;
```

Output



A screenshot of a web browser showing a feedback form at localhost:5173/feedback. The form has a dark background with white text. It includes fields for Name (filled with 'Harsh Shah'), Rating (a dropdown menu showing '4'), Interested in buying again in future (a checkbox labeled 'Yes' which is checked), and Message (Max 25 characters, filled with 'Your service is good'). A blue 'Submit' button is at the bottom left.



A screenshot of a web browser showing the 'Thank You' page at localhost:5173/thankyou. The page has a dark background. A white box in the center contains the text 'Thank You! for giving feedback' and lists the submitted information: Name: Harsh Shah, Rating: 4, Interested in buying again: Yes, and Message: Your service is good.

2. Create a react form having fields like name (text), division (drop down menu) and message (text-area) using the useRef hook.

StudentData.jsx

```
import { useRef } from "react";
import { useNavigate } from "react-router-dom";
import "../FormData.css";

const StudentData = () => {
  const nameRef = useRef(null);
  const divisionRef = useRef(null);
  const messageRef = useRef(null);
  const navigate = useNavigate();

  const handleSubmit = (e) => {
    e.preventDefault();

    const name = nameRef.current.value;
    const division = divisionRef.current.value;
    const message = messageRef.current.value;

    navigate("/acknowledge", {
      state: {
        name,
        division,
        message,
      },
    });
  };

  return (
    <div>
      <h1>Student Details Form</h1>
      <form onSubmit={handleSubmit}>
        <table className="form-table">
          <tbody>
            <tr>
              <td>
                <label htmlFor="name">Name:</label>
              </td>
              <td>
                <input type="text" id="name" ref={nameRef} />
              </td>
            </tr>
            <tr>
              <td>
                <label htmlFor="division">Division:</label>
              </td>
              <td>
                <select id="division" ref={divisionRef}>
                  <option value="1">Division 1</option>
                  <option value="2">Division 2</option>
                  <option value="3">Division 3</option>
                  <option value="4">Division 4</option>
                  <option value="5">Division 5</option>
                  <option value="6">Division 6</option>
                </select>
              </td>
            </tr>
          </tbody>
        </table>
      </form>
    </div>
  );
};
```

```

        </tr>
        <tr>
            <td>
                <label htmlFor="message">Message:</label>
            </td>
            <td>
                <textarea
                    id="message"
                    ref={messageRef}
                ></textarea>
            </td>
        </tr>
        <tr>
            <td colSpan="2">
                <button type="submit">Submit</button>
            </td>
        </tr>
    </tbody>
</table>
</form>
</div>
);
};

```

```
export default StudentData;
```

Acknowledge.jsx

```

import { useLocation } from "react-router-dom";
import "../ThankYouPage.css";

const Acknowledge = () => {
    const location = useLocation();
    const formData = location.state;

    return (
        <div className="thank-you-container">
            <h2 className="thank-you-heading">Thank You for Your Feedback!</h2>
            <div className="thank-you-details">
                <p>
                    <strong>Name:</strong> {formData.name}
                </p>
                <p>
                    <strong>Division:</strong> {formData.division}
                </p>
                <p>
                    <strong>Message:</strong> {formData.message}
                </p>
            </div>
        </div>
    );
};
export default Acknowledge;

```

FormData.css

```
.form-table {  
  width: 100%;  
  border-collapse: collapse;  
}  
  
.form-table td {  
  padding: 8px;  
}  
  
.form-table label {  
  font-weight: bold;  
}  
  
.form-table input[type="text"],  
.form-table select,  
.form-table textarea {  
  width: 100%;  
  padding: 8px;  
  border: 1px solid #ccc;  
  border-radius: 4px;  
  box-sizing: border-box;  
}  
  
.form-table button {  
  padding: 8px 16px;  
  border: none;  
  border-radius: 4px;  
  background-color: #646cff;  
  color: #fff;  
  cursor: pointer;  
}  
  
.form-table button:hover {  
  background-color: #535bf2;  
}
```

ThankYouPage.css

```
.thank-you-container {  
  max-width: 600px;  
  margin: 0 auto;  
  padding: 20px;  
  background-color: #f9f9f9;  
  border-radius: 8px;  
  box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);  
  color: black;  
}  
  
.thank-you-heading {  
  text-align: center;  
  color: black;  
}  
  
.thank-you-details {  
  margin-top: 20px;
```

20CP314P

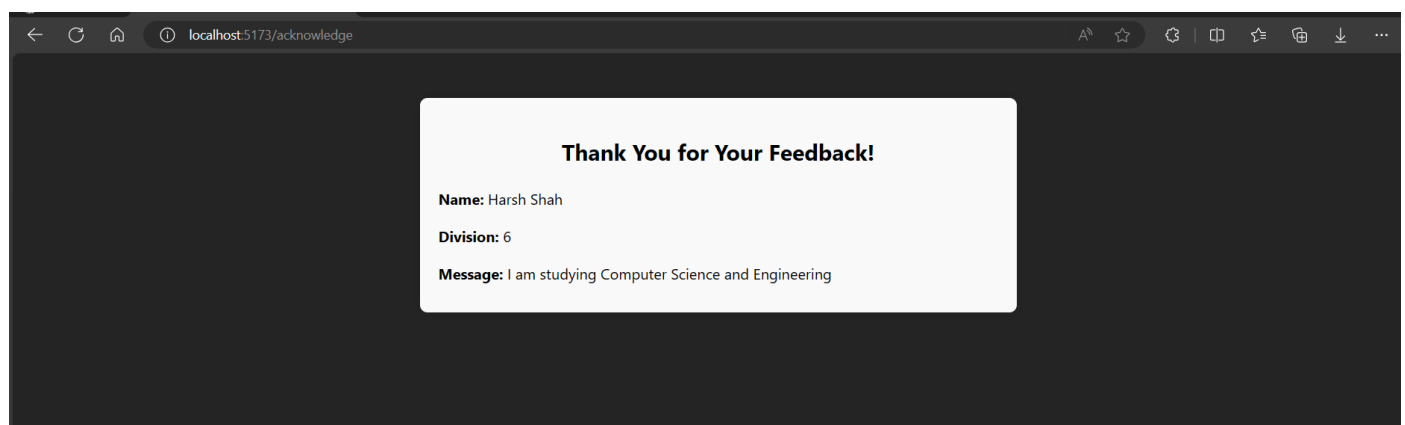
21BCP359

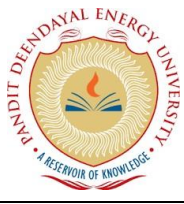
}

```
.thank-you-details p {  
  margin-bottom: 10px;  
}
```

```
.thank-you-details p strong {  
  font-weight: bold;  
}
```

Output



	Course Name: Advanced Web Technology	EXPERIMENT NO. 9
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Working with React hooks

1. Write react programs displaying the usage of each of the below given react hooks:

a. `useReducer`

UseReducer.jsx

```
import { useReducer } from "react";
import { ACTION_TYPES } from "../postActionTypes";
import { INITIAL_STATE, postReducer } from "../postReducer";

const Post = () => {
  const [state, dispatch] = useReducer(postReducer, INITIAL_STATE);

  const handleFetch = () => {
    dispatch({ type: ACTION_TYPES.FETCH_START });
    fetch("https://randomuser.me/api/")
      .then((res) => {
        return res.json();
      })
      .then((data) => {
        const firstName = data.results[0].name.first;
        const lastName = data.results[0].name.last;

        const greet = { message: "Hi " + firstName + " " + lastName };

        dispatch({ type: ACTION_TYPES.FETCH_SUCCESS, payload: greet });
      })
      .catch(() => {
        dispatch({ type: ACTION_TYPES.FETCH_ERROR });
      });
  };

  return (
    <div>
      <h1>useReducer</h1>
      <button onClick={handleFetch}>
        {state.loading ? "Wait..." : "Fetch the post"}
      </button>
      <p>{state.post?.message}</p>
      <span>{state.error && "Something went wrong!"}</span>
    </div>
  );
};

export default Post;
```

postreducer.js

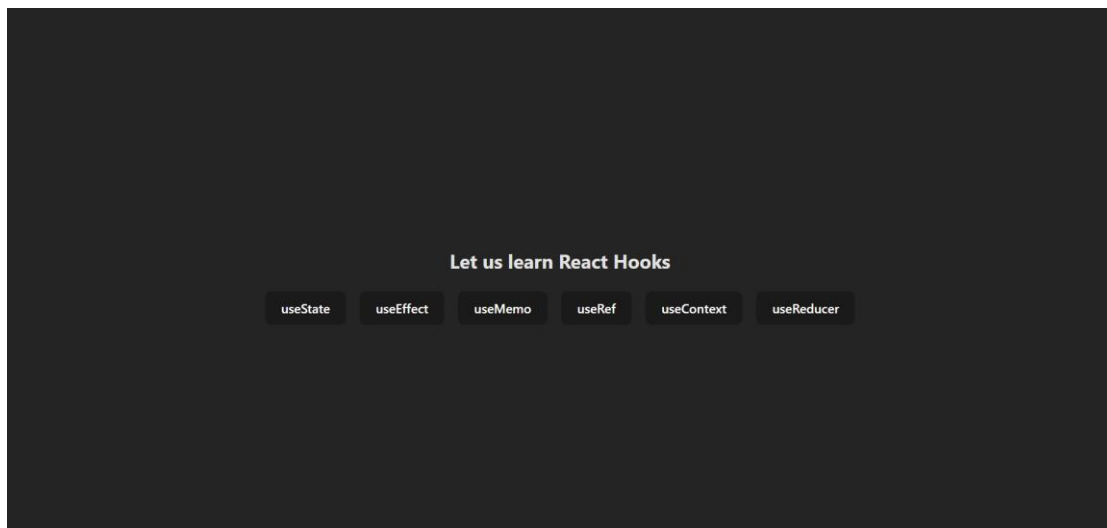
```
import { ACTION_TYPES } from "../postActionTypes";

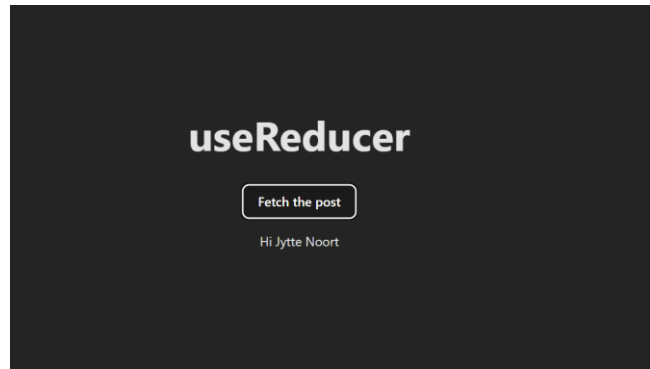
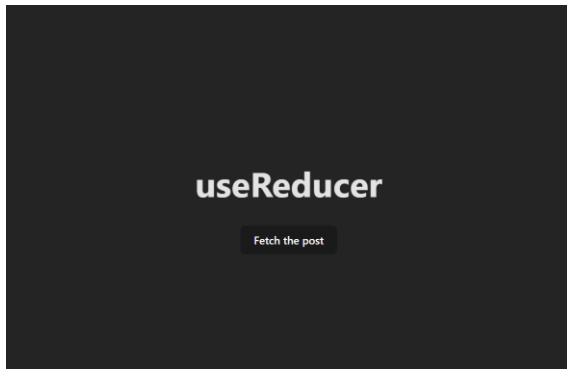
export const INITIAL_STATE = {
  loading: false,
  post: {},
  error: false,
};

export const postReducer = (state, action) => {
  switch (action.type) {
    case ACTION_TYPES.FETCH_START:
      return {
        loading: true,
        error: false,
        post: {},
      };
    case ACTION_TYPES.FETCH_SUCCESS:
      return {
        ...state,
        loading: false,
        post: action.payload,
      };
    case ACTION_TYPES.FETCH_ERROR:
      return {
        error: true,
        loading: false,
        post: {},
      };
    default:
      return state;
  }
};
```

postActionTypes.js

```
export const ACTION_TYPES = {
  FETCH_START: "FETCH_START",
  FETCH_SUCCESS: "FETCH_SUCCESS",
  FETCH_ERROR: "FETCH_ERROR",
};
```





b. useMemo

```
import { useState, useMemo } from "react";
import "../App.css";
function UseMemo() {
  const [count, setCount] = useState(0);
  const [darkTheme, setDarkTheme] = useState(false);
  const squareResult = useMemo(() => {
    delayFunction(count);
  }, [count]);
  const themeStyles = useMemo(() => {
    return {
      backgroundColor: darkTheme ? "blue" : "red",
      color: darkTheme ? "red" : "blue",
    };
  }, [darkTheme]);

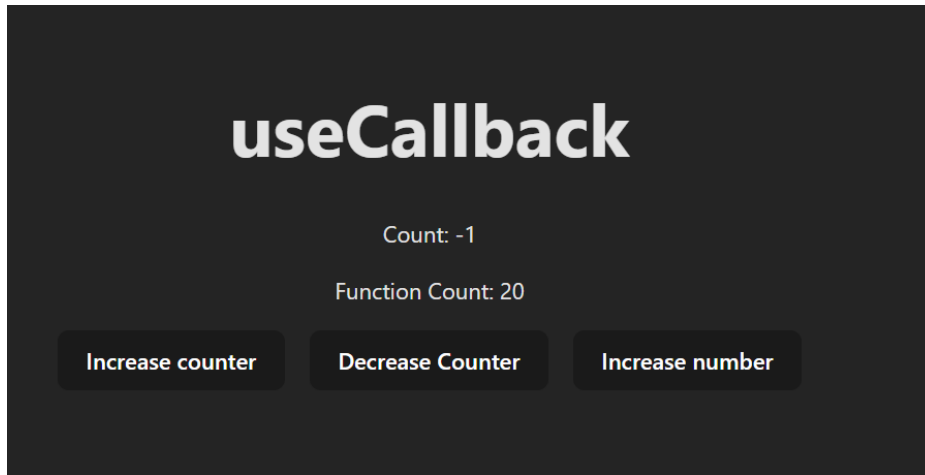
  return (
    <>
      <div className="card">
        <h1>useMemo</h1>
        <label>
          Number:{" "}
          <input
            type="number"
            value={count}
            onChange={(e) => setCount(parseInt(e.target.value))}
          />
        </label>

        <p>Square: {squareResult}</p>

        <button onClick={() => setDarkTheme((prevTheme) => !prevTheme)}>
          Change Theme
        </button>
        <p style={themeStyles}>Theme div</p>
      </div>
    </>
  );
}

function delayFunction(n) {
  for (let i = 0; i < 1000000000; i++);
  return n * n;
}
export default UseMemo;
```





d. useContext

```
import { useState, createContext, useContext } from "react";
```

```
const UserContext = createContext();
```

```
function useContext() {
  const [user] = useState("AWT Student");
```

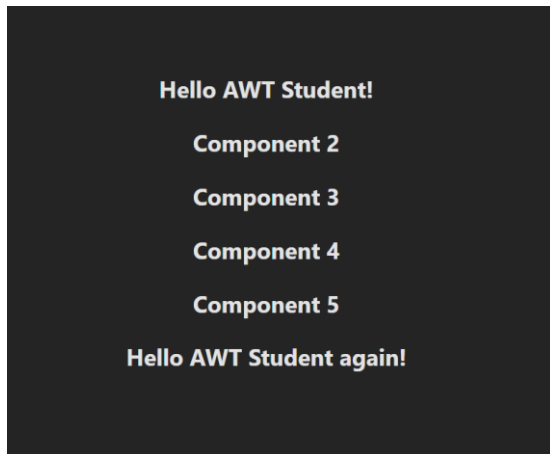
```
  return (
    <UserContext.Provider value={user}>
      <h2>{`Hello ${user}!`} </h2>
      <Component2 />
    </UserContext.Provider>
  );
}
```

```
function Component2() {
  return (
    <>
      <h2>Component 2</h2>
      <Component3 />
    </>
  );
}
```

```
function Component3() {
  return (
    <>
      <h2>Component 3</h2>
      <Component4 />
    </>
  );
}
```

```
function Component4() {
  return (
    <>
      <h2>Component 4</h2>
      <Component5 />
    </>
  );
}
```

```
function Component5() {
  const user = useContext(UserContext);
  return (
    <>
      <h2>Component 5</h2>
      <h2>`Hello ${user} again!`</h2>
    </>
  );
}
export default UseContext;
```



e. useEffect

```
import { useEffect, useState } from "react";
import "../App.css";
function UseEffect() {
  const [count, setCount] = useState(0);
  const [numberProperty, setNumberProperty] = useState("");

  useEffect(() => {
    if (count % 2 === 0) {
      setNumberProperty(`Even Number: ${count}`);
    } else {
      setNumberProperty(`Odd Number: ${count}`);
    }
    console.log("useEffect changed");
  }, [count]);

  return (
    <>
      <div className="card">
        <h1>useEffect</h1>
        <button onClick={() => setCount((count) => count + 1)}>
          Click Me!
        </button>
        <p>{numberProperty}</p>
      </div>
    </>
  );
}
export default UseEffect;
```



2. Design a custom hook that helps you declare a default value of a component

Customhook.jsx

```
import useFetch from "./useFetch";
const Customhook = () => {
  const data = useFetch("https://jsonplaceholder.typicode.com/todos");
  return (
    <div className="card">
      <h1>Custom Hook</h1>
      {data &&
        data.map((item) => {
          return <p key={item.id}>{item.title}</p>;
        })
      }
    </div>
  );
};
export default Customhook;
```

useFetch.js

```
import { useState, useEffect } from "react";
const useFetch = (url) => {
  const [data, setData] = useState(null);

  useEffect(() => {
    fetch(url)
      .then((res) => res.json())
      .then((data) => setData(data));
  }, [url]);

  return data;
};
export default useFetch;
```

